PHYS 2320 Level 2 Coding Project(L2CP)

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Is there a project plan that covers the full length of the project and has milestones with timescales. Are there regular progress reports that reference the plan and indicate the plan is being followed or adapted.

Os I II(i) **II(ii)** III Fail

Based on the score in the autograder output and adjusted for whether answers are correct or not and whether the correct plots have been produced.

Os I II(i) II(ii) **III** Fail

Does the code make use of well designed functions to carry out specific parts of the task. Does it make apporpriate use of loops and data structures to store intermediate results.

Os I II(i) **II(ii)** III Fail

Is the code written to good Python programming standards with repect to identifier names, docstrings and comments, efficient use of data structures and code. Use notes on the autograder output and relative timings to suggest areas to consider.

Os **I** II(i) II(ii) III Fail

Does the code show evidence of awareness of assumptions about inputs and expected outputs and of operations that are prone to failure. Are these handled appropriately with explanative exceptions or warnings.

Os I II(i) II(ii) **III** **Fail**

Are the results reported correctly in section A - correct graphs and appropriate presentation in tabular section (erros to 1s.f. answers rounded to same precision). Is the flow diagram in section B a good description of the code submitted - are structures such as loops, branching on conditions, functiona calls clearly identifiable.

Os **I** **II(i)** II(ii) III Fail

Positive aspects about the code as drawn from the marking criteria. For each criteria, why is this not a fail?

Project Planning and Management: A lot of reports, a long term plan esdtablised from week 1 that is well detailed and realistc. Good reference to details of code in progress reports and what is specifically being worked on. Weekly monotoring against plan. I have raised your mark above 3rd for the reports being very good quality.

Functionality and Implementation of Features: the base mark for the features score is only damped data - 2ii./ Code finds student data file solution and errors correclty (A from eq 8.1 , rest from curvefit)

Structure of Code: Sensible functions that do logical tasks, process data is easy to follow.

Coding Style: Sensible imports, variable names and function names. Very good docstrings., and very good use of comments that helkps someone understand your code easily.

Robustness and Error Handling: one raise statement to check input data.flow chart is well detailed and describes code structure well;.

Code Documentation: Plots have correct axis labels and units results are displayed to correct precision which is same as error which is 1 s.f.

What are the aspects of the code limiting the overall mark. For each criteiria, why is this not 1st/outstanding?

Project Planning and Management: 6 or fewer entries is a 3rd.

Functionality and Implementation of Features: Code crashes on curvefit attempt on assessment data file.

Structure of Code: Code structure is very confusing and functions called iwthin other functions, for example Asymtmetry function calls lots of other functions and is also called in process data. , read_data function is called twice in 2 different codes when it sohuld juist be called once in process data - Ultimately this is making the code take very long to run (3x longer than expected). For Os, code makes use of functions grouped in classes or modules to make reuse and debugging easier.

Coding Style: -1 grade for slow code as discussed in structure section.

Robustness and Error Handling: The raise statement only raises exception, is too broad and doesnt inform user of particular problems.

Code fails to detect the error for the standard data set. Code has very little error handling - these would be sanity checks on input data e.g. is file? is 2 columns? and and checking the output data is valid e.g. are values all floats, not NaNs, in expected range? as well as checks within the code to help with debugging e.g. are curvefits returning sensible values? near p0? e.g. These would be in the form of try/except statements, assertions, raise clauses etc... to catch errors and tell the user the exact problem.

Code Documentation: Results are formatted incorreclty, should be in microtesla and microseconds, you have put in tesla and seconds which makes it very hard to read the results. -1 grade. The flow chart is hard to follow, partly bercause you calkl functions within functions as discussed earlier which confuses the entire call sequence. If everything had been called sequentially from process data, the flow chart would not split immediatley into two seperate branches. Also would aid visually to change boxes to different identifiers for function calls, returns, if/else, try/except,. plots in forms of different box shapes/colours.